

A Cartesian dream:
A geometrical account of syntax
In honor of Andrea Moro

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Guglielmo Cinque

Luigi Rizzi

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A CARTESIAN DREAM

A geometrical account of syntax

In honor of Andrea Moro

Edited by

Matteo Greco & Davide Mocchi



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To our *Maestro*
(Matteo and Davide)

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Introduction

Matteo Greco & Davide Mocci

IUSS PAVIA

La novità del suono e 'l grande lume
di lor cagion m'accesero un disio
mai non sentito di cotanto acume
(Dante, *Comedia*, Paradiso I 82-84)¹

The newness of the sound and the great light
kindled in me a longing for their cause,
never before with such acuteness felt.
(tr. Longfellow 1867)

This tercet from *La Divina Commedia* seems an apt and concise description of Andrea Moro's attitude toward the activity of research. Dante Alighieri is here describing his feeling when he finally reaches Paradise: the sound of celestial spheres and the vision of a dazzling sun burning the sky with its flame sparks the desire to know what causes these phenomena with a previously-unknown degree of intensity. Everyone who has met Andrea in person or through his many books and papers knows that such a desire is what animates his life in several ways and in several fields, particularly with regards to linguistic studies.

1. See Chiavacci Leonardi 1994

1 A Cartesian dream

One way of viewing Andrea's work is as a substantial contribution to Cartesian linguistics, the central doctrine of which is that "the general features of grammatical structure are common to all languages and reflect certain fundamental properties of the mind" (Chomsky 2009: p. 98). Not only does his work help us to better understand the "general features of grammatical structure"; it also provides a precise characterization of the relationship between such features and the "fundamental properties of the mind" - now to be understood in neurobiological terms. Thus, rather than attempting a summary of all of Andrea's linguistic and neurolinguistic works, in what follows we shall briefly illustrate some aspects of his contribution to Cartesian linguistics by drawing connections between his seminal ideas and the crucial properties of Cartesian linguistics as they appear in Chomsky's (2009) text.²

1.1 The creative aspect of language use

The first property of Cartesian linguistics which we would like to focus on is the "creative aspect" of language use, i.e., in the words of Wilhelm von Humboldt, the fact that human language "makes infinite use of finite means." Human language is free from control by external factors, generating thought and sentences that can be shared (or not) with other people, serving as an instrument for free expression rather than as a purely communicative device (Chomsky 2009: p. 76). This aspect emerges in several of Andrea's works, becoming the cornerstone of his approach to linguistics (see, among many others, Moro 2016a): he emphasizes the infinite iteration of syntactic procedures as the fundamental property distinguishing between human and animal languages and between possible and impossible languages. Accordingly, an impossible language is one where the position of a word in the linear sequence is the source of dependencies, whereas a possible language is one where the hierarchical recursive structure is the

2. We refer the reader interested in a systematic summary of Andrea's works up to 2013 to the introduction to Moro 2013.

source of dependencies. In fact, all rules in all attested and existing languages fall into the latter class (Moro 2016b).³

1.2 Deep and surface structure

The second important property of Cartesian linguistics is the distinction between deep and surface structure: “The underlying organization of a sentence relevant to semantic interpretation is not necessarily revealed by the actual arrangement and phrasing of its given components.” (Chomsky 2009: p. 79). As an illustration of this point, consider the famous example from Port-Royal Grammar (Arnauld & Lancelot 1975) “Invisible God created the visible world”, which is taken to be made up of three silent propositions: (i) God is invisible, (ii) He created the world, and (iii) the world is visible. There are, therefore, two different levels: a superficial one - the actually pronounced sound sequence - that contains a perceivable, complex proposition “Invisible God created the visible world”, and a deep one - a mental reality devoted to conveying the meaning of the sentence (Chomsky 2009: p. 80) - that contains three silent, simple propositions: “God is invisible”, “He created the world”, and “the world is visible”.⁴ Thus, a perceivable, complex phenomenon - namely, the set of affirmations obtaining in “Invisible God created the visible world” - is reduced to the interaction of silent, simple elements, namely the subject (that of which one affirms, such as *God* in “God is invisible”), the predicate (that which is affirmed of the subject, such as *invisible* in “God is invisible”), and the copula (the sign of affirmation, such as *is* in “God is invisible”).⁵ In this way, Cartesian lin-

3. Around the distinction between possible and impossible languages Andrea even built a detective novel: see Moro 2018.

4. These three sentences are all simple in the sense that each of them expresses a single judgement (i.e., the form of thought in which something – e.g., *invisible* – is affirmed of something else – e.g., of *God*); by contrast, “Invisible God created the visible world” is complex in the sense that this proposition simultaneously expresses multiple judgements (e.g., both *invisible* and “created the visible world” are affirmed of *God*). See especially Chomsky 2009: pp. 79–80 in this connection.

5. Note that Port-Royal grammarians reduced any sentence – including those whose predicate is verbal – to the subject-copula-predicate schema. Thus, they took the “true” form of a sentence like “Peter lives”, where the predicate is the lex-

guistics abides by one of the guiding principles of modern science: to “explain what is visible and complicated by means of what is simple and invisible” (Perrin 1913).⁶

This second property of Cartesian linguistics - i.e., the distinction between deep and surface structure, alongside the reduction of the complex to the simple - clearly recurs in Andrea’s work. His influential analysis of copular sentences (Moro 1987, 1988, 1993, 1997b, 2017a,b,c) consists essentially in unifying their variety in a transformational way (see Moro 2013: p. 5). Specifically, he proposes that a single deep structure underlies all sentences of the “DP copula DP” type:

- (1) [TP__ [VP V^o_{copula} [SC DP_{subj} DP_{pred}]]]

From the deep structure in (1), the two surface structures in (2)-(3) are derived transformationally, i.e., via movement: (2a), where the DP subject raises to Spec-TP, is dubbed as “canonical copular sentence”; (3a), where it is instead the predicative DP that raises to Spec-TP, is dubbed as “inverse copular sentence”. Inverse copular sentences include *there*-sentences as well, as in (4a). The structural representations in (2a), (3a), and (4a) are exemplified by the English sentences in (2b), (3b), and (4b), respectively.

- (2) a. [TP DP_{subj} [VP V^o_{copula} [SC *t* DP_{pred}]]]
 b. A picture of the wall is the cause of the riot.
- (3) a. [TP DP_{pred} [VP V^o_{copula} [SC DP_{subj} *t*]]]
 b. The cause of the riot is a picture of the wall.

ical verb *lives*, to be “Peter is living” (Chomsky 2009: p. 86; Moro 2017a: p. 42).
 6. See Moro 2017a: p. 41. Incidentally, Andrea’s neurolinguistic work has clarified one of the senses in which a linguistic expression may be silent. Thus, he and his colleagues showed for the first time in Magrassi et al. 2015 that acoustic information is also represented in non-acoustic areas (such as Broca’s area) during endophasic activity – i.e., the ability to express thoughts in verbal form without producing any sound.

- (4) a. $[_{TP} \textit{there}_i [_{VP} V^{\circ}_{\text{copula}} [_{SC} \textit{DP}_{\text{subj}} t]]]$
 b. There is a picture of the wall.

The discovery of inverse copular sentences was made possible by two of Andrea's simple, but revolutionary ideas: (i) that Spec-TP - which had been invariably associated with the subject in the generative tradition - may be occupied by a predicate, while the subject is contained within the verb phrase; (ii) and that *there* is an expletive of the predicate (rather than of the subject).⁷

In this way, the taxonomy of copular sentences (i.e., the surface structures of the form "DP copula DP") reduces to two basic types, namely canonical and inverse copular sentences, which are however the manifestations of a single deep structure. Moreover, the interaction of independently assumed general principles with the structural representations in (2)-(4) captures a series of anomalies for which no explanation had been available in the standard generative model. In brief, then, Andrea's analysis of copular sentences capitalizes on the distinction between deep structure ((1)) and surface structure ((2)-(4)) to decompose a complex phenomenon into the interaction of simple elements.

1.3 The universality of deep structure

The distinction between deep and surface structure is closely related to the third fundamental property of Cartesian linguistics: the universality of deep structure, i.e., the fact that the general features of grammar are common to all languages, linguistic variation being confined to surface structure (Chomsky 2009: p. 81). The question here is what these general features of grammar are. In our opinion, Andrea's contribution to Cartesian linguistics emerges most clearly with respect to this question; accordingly, we shall examine his ideas in greater detail in the present subsection.

As we saw in §1.2 above, Port-Royal grammarians took deep structure - and therefore the general features of grammar common

7. Andrea's idea that *there* is an expletive of the predicate was adopted, with interesting results, in Hale and Keyser's (2002: pp. 189–203) restrictive theory of argument structure.

to all languages - to consist of propositions like “God is invisible” which contain only three ingredients (the subject, the predicate, and the sign of affirmation). Thus, for Port-Royal grammarians the search for the general features of grammar common to all languages essentially boiled down to the search for the “true” form of propositions (the ingredients which a proposition must contain in order to count as a possible proposition). With the advent of generative linguistics, the question of the general features of grammar common to all languages is seen in a new light: the emphasis shifts from the form of propositions to the form of grammatical rules (Chomsky 2009: p. 97). Put another way, generative grammarians search for the formal conditions that any grammar must satisfy in order to count as a “possible” grammar - where “grammar” means a function that takes as input a set of primitive elements of a language *L* (e.g., words) and yields as output only the well-formed sentences of *L* (Chomsky & Moro 2022: p. 84). We turn to such formal conditions directly.

Research in the generative tradition has identified two properties as ubiquitous in human language: hierarchy - i.e., the fact that linguistic units are not on an equal footing - and locality - a filter which blocks some otherwise possible syntactic dependencies.⁸ Thus, any possible grammar must have rules that capture both hierarchy and locality. Interestingly, hierarchy and locality can be captured in configurational terms, i.e., by making reference to the syntactic configurations created by assembling lexical items. Consider for instance long-distance agreement as in (5) - a traditional example of hierarchy-based dependency.⁹

(5) [_{NP} the cat_i that was chased by [_{NP} the dogs]] [_{VP} is_i old].

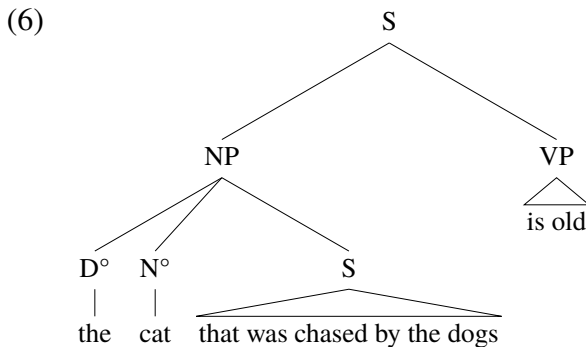
The point is why *is*_[+SING] agrees in number with *cat*_[+SING] but not with *dogs*_[+PL] in (5): if *cat* and *dogs* were on an equal footing, they should behave alike with respect to *is*. Indeed, English words are grouped under “nodes”, i.e., under symbols that are not English

8. The pair formed by these properties is sometimes referred to as “structure dependence” (Moro 2015: p. 89).

9. For ease of exposition, in discussing (5) we shall refer to phrases like “the dogs” as NPs rather than as DPs.

words: e.g., noun phrases (NP) (i.e., the group of words headed by either *cat* or *dogs*) and verbal phrases (VP) (i.e., the group of words headed by *is*). Crucially, phrases can contain other phrases: for instance, the NP headed by *cat* contains the NP headed by *dogs*, as shown in (5). Accordingly, *cat* and *dogs* are not on an equal footing. Now, the relevant rule of verbal agreement is that the verb heading the VP agrees with the noun heading the NP adjacent to VP: thus *is* (i.e., the head of the VP) agrees with *cat* in (5), as opposed to *dogs*, because VP is adjacent to the NP headed by *cat*, but not to the NP headed by *dogs* - in fact, the latter NP is contained within the former NP. Relations between nodes - such as the adjacency relation between NP and VP and the containment relation between the two NPs - are syntactic configurations.

Thus, the hierarchical dimension of language - the fact that linguistic units are not on an equal footing - may be directly expressed in terms of syntactic configurations. But, crucially, syntactic configurations can be given a “geometrical” - i.e., bidimensional - representation by means of a tree diagram (Moro 2015: pp. 61–62): (5) can be represented as in (6), where the relevant relations between nodes are made graphically explicit.¹⁰ Therefore, we conclude that hierarchy can be captured in geometrical terms.

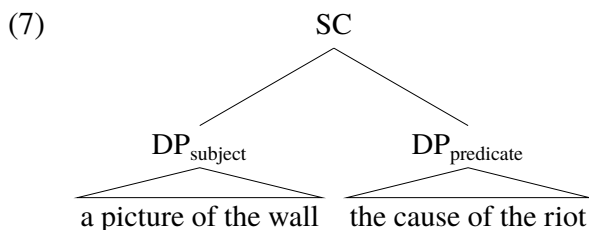


As for the other ubiquitous property in human language, namely locality, it is more controversial whether this can be thoroughly captured

10. (5) is a simplified representation which abstracts away from many irrelevant details.

in geometrical, qua configurational, terms. Nevertheless, a reduction of locality to syntactic configurations has been attempted with very interesting results by Kayne 1984 and this goes under the rubric “Connectedness”.¹¹

Crucially, Andrea exploited the explanatory power of geometrical representations in novel ways, for instance by applying it to predication (i.e., the link between a subject and its predicate). Thus, he conjectured that “all and only predicative structures are codified in syntax by symmetrical constructions” (Moro 2019b: p. 4). Simplifying somewhat, a symmetrical construction is the outcome of merging two units of the same category (e.g., two heads, or two phrases), without the intervention of a head. Thus, consider again sentence (2b) (“A picture of the wall is the cause of the riot”): it involves a predicative link between the DP subject (“a picture of the wall”) and the predicative DP (“the cause of the riot”); in keeping with Andrea’s conjecture, such a link is codified by the symmetrical construction in (7), which is dubbed as “small clause” (SC).



Insofar as (7) is based on purely configurational notions, it is a geometrical representation of predication. Thus, predication can be captured in a geometrical fashion (see Moro 2019b and the references cited therein).

Another linguistic domain for which Andrea has offered a geometrical account is movement. In the canonical theory developed within the Minimalist Program (Chomsky 1995 and much subsequent work), movement is a “pull-chain” phenomenon: simplifying somewhat, a unit *X* moves to a position *Y* because there is a feature *F*

11. We refer to Kayne 1984 for a detailed discussion.

in the vicinity of Y such that F attracts X to Y.¹² In his theory of Dynamic Antisymmetry (Moro 1997a, 2000, 2004, 2009), Andrea argued that movement should instead be regarded as a “push-chain” phenomenon:¹³ movement is not “driven” by features; rather, it is a repair strategy that intervenes to restore asymmetry in symmetrical syntactic structures. Consider how.

Simplifying somewhat, any syntactic structure must be linearized and labeled (Moro & Roberts 2023). The algorithms that provide for linearization and labeling are formulated in such a way as to be satisfied in asymmetrical structures only.¹⁴ As a consequence, symmetrical structures such as the small clause of (7) are non-linearized and unlabeled, and hence, illegitimate. Crucially, movement can restore asymmetry into a symmetrical structure. For instance, the movement of either component of the small clause turns the small clause into an asymmetrical structure, thereby allowing for both linearization (Moro 1997a, 2000, 2004) and labeling (Moro 2009), such as in canonical and inverse copular sentences (see §1.2 above). Andrea’s proposal was to intertwine movement with symmetrical structures, in keeping with the following conjecture: movement is driven by the search for asymmetry (Moro 2000: p. 28). In this way, Dynamic Antisymmetry handles movement in purely geometrical terms, insofar as the notion of symmetrical - and asymmetrical - syntactic structure is purely configurational (i.e., is based on relations between nodes), and therefore geometrical.¹⁵

Other domains of syntax can be given a geometrical account. Building on the fact that, among other things, hierarchy, locality, pre-

12. We would like to recall here Andrea’s insightful remarks on the meaning of “minimal” in the Minimalist Program, as well as on the tricky relationship between the Minimalist Program and the notions of competence and performance: see Moro 1996.

13. The use of the terms “pull-chain” and “push-chain” to characterize contemporary approaches to movement is borrowed from Moro & Roberts 2023.

14. See Kayne 1994: pp. 3–6 on the formulation of the linearization algorithm, dubbed as “Linear Correspondence Axiom”, and Chomsky 2013: pp. 43–44 on the formulation of the labeling algorithm.

15. Dynamic Antisymmetry has had a far-reaching impact on contemporary linguistic theory, being in fact at the basis of Chomsky’s formulation of the labeling algorithm (see, among others, Chomsky 2013: p. 43 and Rizzi 2015: p. 21).

dication, and movement can be captured in geometrical terms, Andrea has advanced a conjecture: that linguistic phenomena can all be captured in geometrical terms, i.e., by referring to the syntactic configurations generated by assembling lexical items and represented on a bidimensional space. He uses to call such a conjecture “a Cartesian dream”, where the adjective “Cartesian” conveys the properties of Cartesian linguistics as presented in Chomsky 2009 and discussed in this introduction, but also pays homage to Descartes’s use of a bidimensional space - as defined by the Cartesian coordinates x, y - to translate algebraic expressions into geometrical expressions (and vice versa). The following passage may be taken as a sort of *Manifesto* of the Cartesian dream:

I see two major questions rising above the others. The first, from a formal perspective, is whether all principles of grammar could be captured configurationally: that is to say, whether the geometry of language might not be the actual stuff from which language structure is made. Locality as suggested in the so-called “connectedness theory” proposed by Kayne 1984 [*Connectedness and Binary Branching*, Dordrecht: Foris], semantic role assignment as in Hale and Keyser’s (2002) [*Prolegomenon to a theory of argument structure*, Cambridge, MA: MIT Press] approach, movement as in dynamic antisymmetry (Moro 2000) [*Dynamic Antisymmetry*, Cambridge, MA: MIT Press], the fine-grained map of clause structure as in the “cartographic project” (Cinque and Rizzi 2008) [*The cartography of syntactic structures*, in B. Heine and H. Narrog (eds.), *The Oxford Handbook of Linguistic Analysis*, Oxford: Oxford University Press]: all converge on the possibility of rendering the entire language structure in a purely geometric way. If this approach were carried out in full, impossible languages would be, so to speak, a sort of “non-Euclidean” grammar—a grammar in which the metric fails for lack of some structural postulate. (Moro 2016b: p. 121).

Thus, in accordance with the Cartesian dream, the general features of grammar common to all languages - i.e., the universal deep structure of Cartesian linguistics - are of a geometrical nature: they can be reduced to the relationships between nodes in a bidimensional space.

1.4 The mind-language relationship

The fourth fundamental property of Cartesian linguistics - the last, at least for the purposes of the present introduction - is the link between deep structure and the human mind: Cartesian linguists account for the universality of deep structure (the fact that the general features of grammar are common to all languages) by directly linking it to the fundamental properties of the human mind, which are also universal: such properties - defined by Helbert of Cherbury as “a direct gift of Nature, a precept of natural instinct” (*De Veritate*, quoted in Chomsky 2009: p. 98) - are shared by all people and activated by appropriate experience. Such a link is still established in contemporary linguistics, but on a different ground, i.e., the neurobiological one: the general features of grammar that are common to all languages, such as hierarchy and locality (see §1.3 above), depend on the human brain, which is of course universal. Indeed, the fact that, among the infinitely many grammars that we may think of, only a very small subset thereof is realized across languages (i.e., the grammars that correspond to the possible languages as defined in §1.1 above - Moro 2015: p. 1) would be paradoxical - to quote the famous book of Eric Lenneberg 1967 - if languages were just a matter of arbitrary and cultural convention. However, the existence of a subset of grammars is exactly what we expect if we are endowed with a biologically determined guide to language (Moro 2015: pp. 99–102).¹⁶ Lenneberg discusses some (mostly) clinical observation of pathologies to show that the latter interpretation of language is correct. Andrea’s studies go even further showing that, on the basis of new empirical evidence (Moro 2013; 2015), the syntactic structure is autonomously represented in the brain and that the distinction between possible and im-

16. That clearly does not exclude that some aspects of languages are indeed a matter of arbitrary and cultural convention. See Moro 2015 for a detailed discussion.

possible languages is reflected in the way our brain works: the brain does not consider rules based on linear order as part of the language system, whereas it does consider recursive dependencies, i.e., those based on “infinite use of finite means”, as part of the language system. The boundaries of Babel are therefore not just arbitrary and cultural conventions, but are the expression of the brain, which leads us to “the surprising conclusion that it’s flesh that became logos, not vice versa” (Chomsky & Moro 2022: p. 56). This can also be viewed as a sort of antidote against racism (Moro 2019a): at the relevant level of representation, no languages are “better” than others, and reality cannot be seen differently depending on the specific language spoken.

2 A few notes on this book

This book springs from the idea of celebrating Andrea’s 60th birthday with a collection of striking remarks on linguistics and neurolinguistics. To this end, we asked a number of Andrea’s colleagues to write a very short essay which sheds some light on a single but significant point of the linguistic system. The result has been rewarding in several respects, with the essays spanning a wide range of topics.

Thus, some authors focus on copular sentences (Adriana Belletti; Marcel den Dikken; Carlo Cecchetto and Caterina Donati; Gennaro Chierchia), others on the basic operations of movement (Valentina Bianchi; Cristiano Chesi) and agreement (Alessandra Tomaselli). Some investigate aspects that are central to the theory of Dynamic Antisymmetry, including labeling and linearization (Denis Delfitto; Massimo Piattelli-Palmarini; Robert Frank, William Min, Abhisar Mittal, and Billy Zhong; Maria Rita Manzini; Luigi Rizzi; Ian Roberts). Some authors delve into the grammar of different linguistic registers (Maria Teresa Guasti; Liliane Haegeman). Some authors provide new insights into functional heads in both the clausal and the nominal domain (Guglielmo Cinque; Anna Maria Di Sciullo; Cecilia Poletto). Finally, other authors capitalize on computational linguistics (Robert C. Berwick), cognitive sciences (Andrea Serino), aphasia (Stefano F. Cappa and Eleonora Catricalà), parametric variation (Giuseppe Longobardi and Alessandro Treves), and linguistic

errors as heuristic tools (Valentina Bambini), to illuminate the relationship between neurosciences and human language.

Alongside these short essays, the present collection also features two longer papers by Noam Chomsky and Richard S. Kayne. These authors, who are the *Maestri* - directly or indirectly - of all the distinguished contributors of this volume, deal with “*half, all, both, each and of*” aspects of that “*miracle creed*” instantiated by language.

We are very grateful to all those scholars - both internal and external to this volume - who generously accepted to serve as reviewers. We would like to warmly thank each of them: Víctor Acedo-Matellán, Cynthia Allen, Adriana Belletti, Valentina Bianchi, Theresa Biberauer, Luca Bischetti, Marta Bosia, Veronica Bressan, Stefano Cappa, Eleonora Catricalà, Carlo Cecchetto, Cristiano Chesi, Guglielmo Cinque, Silvio Cruschina, Roberta D'Alessandro, Aniello De Santo, Denis Delfitto, Marcel den Dikken, Anna Maria Di Sciullo, Caterina Donati, Ludovico Franco, Robert Frank, Achille Fusco, Chiara Gianollo, Paolo Lorusso, Maria Rita Manzini, Shota Momma, Luigi Rizzi, Sarah Rossi, Giuseppe Rugna, Leonardo Russo Cardona, Giuseppe Samo, Andrea Serino, Ur Shlonsky, Alessandra Tomaselli, Alfredo Tomasetta, Deirdre Wilson, Roberto Zamparelli, Hedde Zeijlstra, Stanislao Zompì.

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